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Co-design of family interventions in cancer care: A scoping review to assess key methods and processes

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ABSTRACT

Purpose: Cancer affects not only patients but also their families, with family caregivers experiencing heightened stress and health risks. Despite numerous support interventions, family caregivers often report unmet needs, largely due to the limited implementation of these interventions into routine cancer care or their lack of relevance to family needs. This scoping review aimed to systematically map the evidence on co-design approaches in family interventions within oncology, examining key themes, methodologies, and outcomes of interventions.

Methods: A scoping review was conducted. EMBASE, MEDLINE, PsycINFO, Scopus, and CINAHL were searched for original research articles in August 2024, with no date restrictions. Studies on co-designed family interventions in cancer care involving adult patients with cancer and their adult family caregivers were included. Data were analyzed using descriptive content analysis.

Results: Of 3024 references screened, fourteen studies were included. Four themes emerged: (a) Co-design approaches and frameworks, (b) Facilitators in the co-design process, (c) Challenges in the co-design process, and (d) Outcomes of the co-design process. The studies employed various co-design frameworks. Key facilitators of successful co-design included high stakeholder engagement, structured frameworks, a diversity of stakeholders, and training and capacity-building. Common challenges were recruitment difficulties, cultural barriers, and time constraints. The co-designed interventions addressed specific needs, enhancing relevance and effectiveness.

Conclusion: Co-design processes in family interventions in cancer care show promise in developing relevant and effective support strategies. Future research should focus on overcoming identified challenges and improving reporting standards to enhance the transparency and reliability of co-design studies.

1. Introduction

Cancer is a family affair (Coyne et al., 2020), meaning that at times of cancer, the health and well-being of the whole family are at risk. For instance, family caregivers of persons diagnosed with cancer have a higher risk of stress-related illness, i.e., ischemic and coronary heart disease and stroke, compared to the general population (Ji et al., 2012; Möllerberg et al., 2016). Risks that persist over time. For this reason, support for family caregivers is stressed, and numerous support interventions have been developed (Samuelsson et al., 2020; Treanor et al., 2019). Despite that, family caregivers continue to report unmet support needs (Christophe et al., 2022). This motivates continued study on how support can be offered to families affected by cancer.

Suggested reasons for the persistent unmet needs are that these support interventions are not implemented into routine cancer care (Ferrell and Wittenberg, 2017) or are irrelevant to the family caregivers' needs (Samuelsson et al., 2022). As a result, despite showing promising results on family caregivers' health and well-being in research, support interventions may not reach the larger population of family caregivers in need (Ferrell and Wittenberg, 2017). A co-design approach to intervention development is recommended to overcome implementation barriers and enhance content relevance (Skivington et al., 2021).

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Co-design and participatory methods are processes where participants and other relevant stakeholders form a partnership and actively participate in intervention development and dissemination (Eyles et al., 2016). Co-design is a type of participatory research focused on shared decision-making and iterative development, while participatory approaches more broadly vary in stakeholder involvement. These methods ensure interventions reflect the perspectives of those affected. For example, Palmar et al. (2025) describe a co-design process where caregivers, healthcare providers, educators, and policymakers developed an inclusive caregiver education program. Engaging caregivers in identifying challenges made training for healthcare providers more relevant, improving caregiver support and well-being.

Involving end users and receivers of interventions in the research processes, e.g., prioritizing research questions and strategies and timings for data collection (Bailey et al., 2015), is suggested to lead to more relevant research and enhanced uptake (Domecq et al., 2014). In addition, it is suggested to improve recruitment response and retention rates (Molloy et al., 2019). That is why doing research "with" instead of "on" people is endorsed (Bryant et al., 2022; Hall et al., 2021; Oliver et al., 2008; Polanco et al., 2022). Hence, by co-designing interventions, these can be developed to fit prerequisites in routine cancer care yet still be relevant to the family caregivers' needs (Skivington et al., 2021).

Co-design involves a wide spread of activities across the whole research process – from formulating the research question to disseminating findings (Eyles et al., 2016), all with a suggested positive impact on research. While previous reviews have examined family interventions in cancer care, no previous review in the included search languages has systematically mapped the evidence on co-design approaches and activities within these interventions, hence the focus of this review. Additionally, this scoping review will describe the characteristics and contexts of these co-design efforts and identify key themes, methodologies, and outcomes associated with the interventions. Thus, the review will offer an overview of co-design approaches and activities used. It will serve as a foundational step for enhancing our understanding and application of co-design in oncology, guiding future research and practice in developing more effective and collaborative interventions for cancer patients and their families.

2. Aim

The objective of this scoping review was to systematically map the literature to provide an accessible synopsis of current co-design approaches and activities in family interventions in cancer.

3. Methods

This review was reported according to the Preferred Reporting Items for Systematic Reviews and Meta-extension for Scoping Reviews (PRISMA-ScR) checklist (Tricco et al., 2018). A scoping review method was selected to outline different types of evidence on the area of interest and the gaps for further research (Munn et al., 2018). This review was undertaken iteratively following the stages described by the Joanna Briggs Institute (JBI) (Peters et al., 2020): identifying the research question, retrieving relevant studies, selection of studies, charting data, and collating, summarizing and reporting evidence. The review was registered in the Open Science Framework (https://doi.org/10.17605/OSF.IO/2RWB9) and a study protocol was created (https://doi.org/10.1136/bmjopen-2024-093961)

3.1. Stage 1: identifying the research question

The following research questions were identified as the main research questions of this scoping review.

1. What approaches to research co-design exist in family interventions in oncology?
2. What activities are involved in these research co-design approaches within family interventions?
3. What is known about reported hinders and impact of the co-design activities on study outcomes in family interventions?

3.2. Stage 2: identifying relevant studies

The literature search followed the three-step strategy recommended by the Joanna Briggs Institute for conducting a scoping review (Aromataris and Riitano, 2014). In June 2024, an initial search in MEDLINE (Ovid) and EMBASE (Ovid) was carried out to identify relevant keywords found in titles, abstracts, and index terms of pertinent studies. The identified keywords and index terms were utilized to conduct a comprehensive database search. The search strategy was elaborated with the assistance of a librarian with expertise in literature review. The search was piloted by the first and last author, who screened the first 500 references in EMBASE; based on this, we chose to include 'co-jointly' as a search term. The first author conducted the database searches in August 2024 across the following databases: EMBASE (Ovid), MEDLINE (Ovid), PsycINFO (Ovid), Scopus, and CINAHL (EBSCO). No restrictions were placed on publication dates. Finally, a backward citation search and forward citation tracking of reference lists were performed on the included studies to locate relevant studies not identified in the previous phase. The complete search strategy used in EMBASE is described in Table 1. Complete search strategies for the other databases are provided as supplemental material: MEDLINE – Supplemental File 1, PsycINFO – Supplemental File 2, Scopus – Supplemental File 3, and CINAHL – Supplemental File 4. Grey literature was excluded.

3.2.1. Inclusion criteria

The inclusion criteria for this review were established: Studies were considered eligible if they involved adult patients diagnosed with any cancer and their adult family caregivers, with "family" and "family caregivers" used interchangeably. Here, "family" is defined broadly to include any group of two or more individuals connected by strong emotional bonds, a sense of belonging, and a commitment to mutual involvement, regardless of blood or social ties (Shajani and Snell, 2023). Studies involving family caregivers of adults with cancer were considered eligible for inclusion, even if the patient is not involved. Studies describing interventions that incorporate co-design or participatory methods at any stage – development, implementation, or dissemination – will be included (Eyles et al., 2016). In this review, co-design refers to a structured collaboration where patients, family caregivers, healthcare professionals, and/or researchers actively contribute throughout the research process, from setting priorities and designing the study to data analysis and sharing results. Family interventions were defined as any approach aimed at promoting, maintaining, or restoring family health or the health of family caregivers, and this review considered all clearly defined interventions or experimental studies targeting family caregivers, whether or not they were physically present in the intervention (Shajani and Snell, 2023). Eligible study types were qualitative, quantitative, or mixed-methods research that employed co-design to create new family interventions or improve or adapt existing ones within oncology or hematology services, as well as studies evaluating the co-design process. Studies published in English, Danish, Norwegian, Spanish, and Swedish were eligible for inclusion.

3.3. Stage 3: study selection

Search results collected from the electronic databases were exported to Covidence software for processing (Covidence, 2024). The selection of studies was executed in a three-step manner. First, duplicate studies were removed. Second, the first and the last author independently screened titles and abstracts of all references to determine eligibility. Discrepancies were discussed with the second author. In the third step, the first and the last author read studies in their entirety and mirrored

Table 1
| Search string example.

Search profile for EMBASE	
Set	Search Statement
#1	malignant neoplasm/
#2	(cancer* or carcinoma* or malignant neoplas* or malignant neoplas* disease or malignant tumor* or malignant tumour* or neoplas* malignan* or oncologic* malignan* or tumor* malignan* or tumour* malignan*).ti,ab,kf.
#3	hematology/
#4	(hematolog* or haematolog* or hematolog* cancer* or haematolog* cancer*).ti,ab,kf.
#5	hematologic malignancy/
#6	(hematolog* malignanc* or haematolog* malignanc*).ti,ab,kf.
#7	1 or 2 or 3 or 4 or 5 or 6
#8	(co-design or co design or codesign or research co-design or research co design or research codesign or co-opera* design or participa* design or co-crea* or co-produ* or co-joint*).ti,ab,kf.
#9	(patient and publ* involv*).ti,ab,kf.
#10	stakeholder engagement/
#11	(stakehold* engag* or stakehold* involve* or stakehold* participa* or consum* involve* in research).ti,ab,kf.
#12	participatory research/
#13	(participa* research or community-based participa* research).ti,ab,kf.
#14	(patient and stakehold* engag*).ti,ab,kf.
#15	consum* engag*.ti,ab,kf.
#16	participa* method*.ti,ab,kf.
#17	inclu* health research.ti,ab,kf.
#18	communit*-academic* partnership*.ti,ab,kf.
#19	communit*-based participa* research.ti,ab,kf.
#20	patient engagement/
#21	(patient engag* or famil* engag*).ti,ab,kf.
#22	8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21
#23	family therapy/
#24	(famil* therap* or famil* interven* or famil* treatment*).ti,ab,kf.
#25	famil*-based interven*.ti,ab,kf.
#26	famil* educa*.ti,ab,kf.
#27	caregiver/
#28	caregiver*.ti,ab,kf.
#29	famil* caregiver*.ti,ab,kf.
#30	carer*.ti,ab,kf.
#31	famil* carer*.ti,ab,kf.
#32	significant other.ti,ab,kf.
#33	next of kin.ti,ab,kf.
#34	relative*.ti,ab,kf.
#35	family/
#36	famil*.ti,ab,kf.
#37	#23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35 or #36
#38	7 and 22 and 37
#39	#38 and Filters: No restrictions were placed on publication dates. Languages included: English, Spanish, Swedish, Norwegian, and Danish

*Note:/*indicates subject headings searches, * indicates that all types of endings are included in the search.

them against the predefined inclusion criteria. The second author resolved disagreements.

3.4. Stage 4: charting the data

A customized data extraction template was developed in Covidence. The template captured general data such as title, authors, year of publication, country, and study aim. Furthermore, the template captured data on the co-design approach, methods, analysis, stakeholder characteristics, co-design facilitators and hinders, and the family intervention. All authors carried out the data extraction. Additionally, the Joanna Briggs Institute (JBI) Critical Appraisal Tools were used to evaluate the quality of included studies. All articles were included in the scoping review regardless of their results, as the primary aim is to gather all available evidence. Following this, descriptive evidence mapping was conducted.

3.5. Stage 5: collating, summarizing, and reporting the results

In line with the methodology from JBI (Peters et al., 2020), descriptive content analysis was applied to the extracted data. The results were mapped using a table to outline data emphasizing key study characteristics and research aims. To ensure clarity in categorizing and synthesizing details on co-design and participatory methods, the results were systematically classified by co-design approach, methods, and stakeholders' characteristics, including their preparation for participation in the co-design process. A detailed narrative report provided more profound insights into the findings, a synthesis of the results and a critical discussion of their implications. All authors participated in the analysis and reporting of the results.

4. Results

4.1. Search results

After removing duplicates, 3024 references were screened for eligibility, and 14 studies met the inclusion criteria and were included in this review (see PRISMA Flow Diagram in Fig. 1). These studies were conducted in Australia (n = 6) (Coyne et al., 2022; Hiatt et al., 2023; Hyatt et al., 2021; Lethborg et al., 2022; Loeliger et al., 2023; White et al., 2021), the United Kingdom (n = 4) (Marshall-McKenna et al., 2023; Milton et al., 2022; Santin et al., 2023; Santin et al., 2019), the United States (n = 2) (Kamen et al., 2022; Pozzar et al., 2022), Norway (n = 1) (Bergerød et al., 2021), and Vietnam (n = 1) (Santin et al., 2020).

The studies included various participant groups: five studies included patients, family caregivers, and healthcare professionals (Coyne et al., 2022; Hiatt et al., 2023; Loeliger et al., 2023; Marshall-McKenna et al., 2023; Pozzar et al., 2022); three studies included family caregivers and healthcare professionals (Bergerød et al., 2021; Santin et al., 2020, 2023); two studies included patients, family caregivers, and either patient advocates or charity representatives (Kamen et al., 2022; Milton et al., 2022); and another two studies included patients, family caregivers, healthcare professionals, and consumer advocates (Hyatt et al., 2021; Santin et al., 2019). One study included both professionals and community members (Lethborg et al., 2022), and another included only family caregivers (White et al., 2021).

In total, the studies included n = 143 patients with cancer, n = 144 family caregivers, n = 258 healthcare professionals, and n = 11 patient/consumer advocates or charity representatives. One study had missing data on n = 12 stakeholders (Santin et al., 2019), and four studies did not specify the number or type of stakeholders who participated in the co-design process (Coyne et al., 2022; Lethborg et al., 2022; Pozzar et al., 2022; Santin et al., 2023) (see Table 2).

4.2. Critical appraisal of included studies

JBI's Critical Appraisal Tool for Qualitative Research was used to assess the quality of the included studies (Table 3) (Lockwood et al., 2015). The appraisal revealed several areas of concern. The included studies failed to address the relationship between the researchers and participants (Q6), and the researcher's influence on the research and vice versa (Q7). In more than half of the included studies, it is unclear whether the research methodology is appropriate to address the research question (Q1).

Despite these shortcomings, the studies demonstrated strengths in other areas, such as robust participant engagement (Q8) and articulation of the co-design process (Q3).

4.3. Synthesis and reporting of study themes

Four themes emerged from the data: (a) Co-design approaches and frameworks, (b) Facilitators in the co-design process, (c) Challenges in the co-design process, and (d) Outcomes of the co-design process.

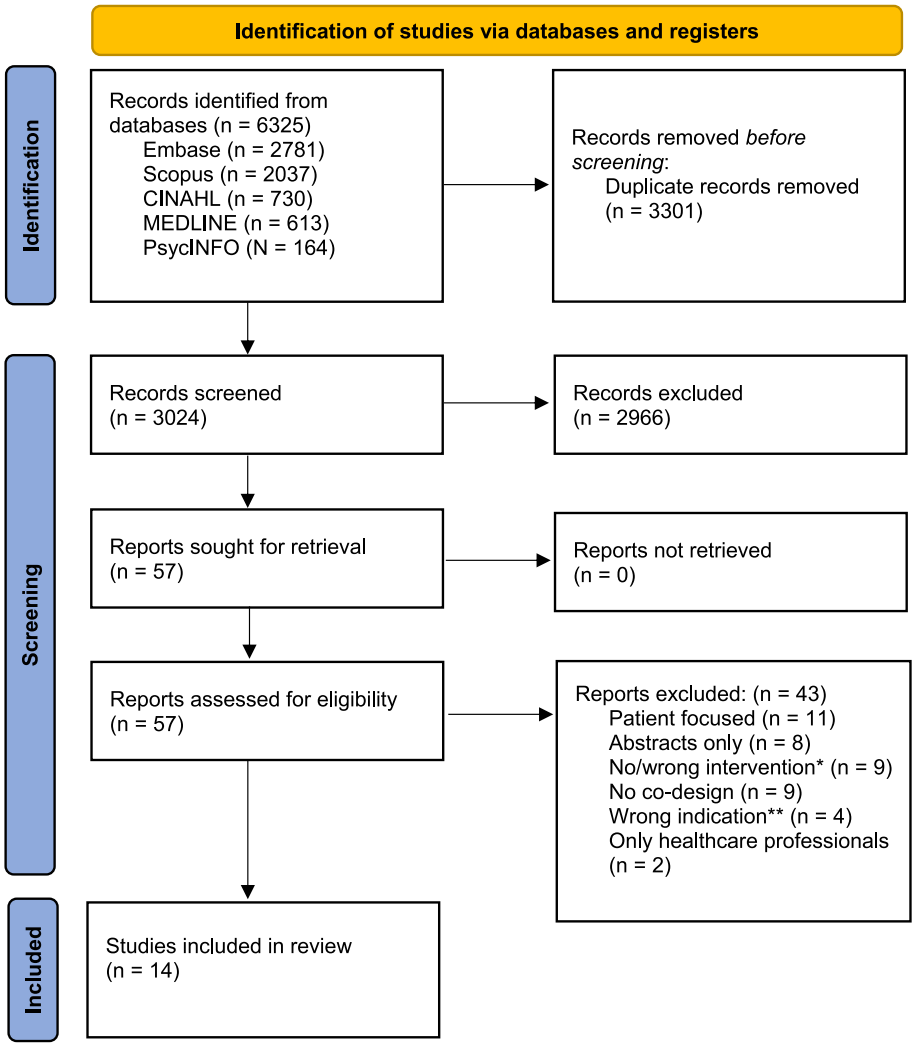


Fig. 1. | PRISMA Flow Diagram of study selection
*Wrong intervention: in this review 'wrong intervention' was identified as an intervention focused on educating healthcare professionals, not patients and family caregivers.
**Wrong indication: in this review 'wrong indication' refers to publications focusing on the effectiveness of patient and public involvement in research, not how the stakeholders participate in co-design activities to develop family interventions in cancer care.

4.3.1. *Co-design approaches and frameworks*

The studies employed various co-design approaches; four studies used the Experience-based co-design approach (Coyne et al., 2022; Hiatt et al., 2023; Hyatt et al., 2021; Loeliger et al., 2023), three studies used the six-step co-design approach described by Santin et al. (2019) (Santin et al., 2019, 2020, 2023), and two used the ADAPT-ITT framework (Kamen et al., 2022; Pozzar et al., 2022). Additionally, one study used an appreciative inquiry approach (Lethborg et al., 2022), another used the Framework for the Co-production and Prototyping of Public Health Interventions (Milton et al., 2022), and one study used a consensus process (Bergerød et al., 2021). One study did not specify the applied co-design approach (White et al., 2021), while another employed a multi-methods study to assess needs, preferences, and expectations in supportive cancer care (Marshall-McKenna et al., 2023).

Among the most commonly used co-design activities were interviews (Coyne et al., 2022; Hiatt et al., 2023; Hyatt et al., 2021; Marshall-McKenna et al., 2023; Milton et al., 2022; Pozzar et al., 2022; Santin et al., 2020; White et al., 2021), workshops (Hiatt et al., 2023; Hyatt et al., 2021; Milton et al., 2022; Santin et al., 2019, 2023), focus groups (Kamen et al., 2022; Loeliger et al., 2023; Santin et al., 2020), and feedback/verification sessions (Bergerød et al., 2021; Hiatt et al., 2023; Hyatt et al., 2021; Kamen et al., 2022; Lethborg et al., 2022; Pozzar et al., 2022; Santin et al., 2020, 2023).

4.3.2. *Facilitators in the co-design process*

Based on the included studies, reported key factors of successful co-design in family interventions for cancer care are a high degree of participant engagement, structured frameworks, a diversity of stakeholders, and training and capacity-building. Hiatt et al. and Loeliger et al. highlighted the enthusiasm and active involvement of patients, family caregivers, and healthcare professionals as crucial to the co-design process (Hiatt et al., 2023; Loeliger et al., 2023). Active participation from and inclusion of a diverse group of stakeholders ensured that the specific needs and preferences of the stakeholders were addressed, making the co-designed family intervention more relevant and useable, potentially leading to better adherence to the intervention (Hiatt et al., 2023; Loeliger et al., 2023). Structured frameworks and toolkits also played a significant role in guiding the co-design activities. Hiatt et al. found that referring to the Experience-Based Co-Design (EBCD) toolkit and case examples was particularly helpful in structuring their workshops and feedback events (Hiatt et al., 2023). Similarly, Milton et al. emphasized the importance of using the Framework for the Co-production and Prototyping of Public Health Interventions, which provided a clear framework for engaging stakeholders in the design

Table 2
| Study characteristics of n = 14 studies.

First author, publication year, location	Aims	Study type	Co-design approach, methods, and preparation of stakeholders	Analysis	Stakeholder characteristics	Co-design facilitators and challenges	Intervention	Key outcomes
Hiatt et al. (2023), Australia	To co-design a framework to improve access to nutrition information and support for patients and carers with head and neck cancer	Qualitative	Experience-based co-design (EBCD) (Dawda and Knight, 2024). Semi-structured interviews, feedback events, co-design workshops. Preparation in EBCD toolkit and case examples.	Interviews were analyzed using reflexive thematic analysis (Braun and Clarke, 2019).	n = 10 patients with head and neck cancer, n = 7 carers of patients with head and neck cancer, n = 15 healthcare professionals.	*Facilitators:* High engagement, useful toolkit *Challenges:* Scheduling issues, low participation in events	Framework for nutrition information and support	Improved access to tailored nutrition information
Loeliger et al. (2023), Australia	To co-design a cancer nutrition care pathway to guide and improve the provision of consistent, evidence-based care	Mixed-methods	Experience-based co-design (EBCD). Literature review, survey, focus groups, co-design workshops, feedback survey. Preparation in EBCD methodology.	Not specified	n = 10 patients, n = 4 carers, n = 50 healthcare professionals.	*Facilitators:* Divers cohort, high engagement *Challenges:* Time constraints, skewed views due to majority healthcare professionals	CanEAT pathway for cancer nutrition care	Centralized suite of resources, tools, and clinical guidance
Marshall-McKenna et al. (2023), UK	To evaluate healthcare needs, preferences, and expectations in supportive cancer care	Mixed-methods	Descriptive, cross-sectional, multi-methods study. Surveys, qualitative interviews.	Descriptive statistics, framework analysis	n = 70 cancer survivors, n = 23 family caregivers, n = 62 healthcare professionals.	*Facilitators:* Stakeholder co-creation insights. *Challenges:* Difficult to engage family caregivers.	LifeChamps digital platform	Enhanced supportive cancer care for older adults
Santin et al. (2023), UK	To outline the rapid co-design process and content of a COVID-19 support and information module	Qualitative	Six-step co-design model (Santin et al., 2019). Online co-design workshops, group feedback sessions.	Thematic analysis	n = 45 stakeholders (family caregivers and healthcare professionals)	*Facilitators:* Iterative framework, rapid development. *Challenges:* Difficult to recruit family caregivers.	COVID-19 support and information module	Video and written information for family caregivers
Coyne et al. (2022), Australia	To develop and evaluate video resources that provide health information for persons caring for someone with brain or general cancer	Mixed-methods	Experience-based co-design (EBCD). Literature review, personal and semi-structured interviews, online survey, Youtube data analytics.	Content analysis, Youtube data analytics	Researches, clinicians, patients, and family caregivers	*Facilitators:* Tailored resources to consumers needs *Challenges:* Not reported	Video resources for family caregivers	Reduced burden of caregiving
Kamen et al. (2022), USA	To use the ADAPT-ITT model to guide the identification and adaptation of evidence-based intervention for LGBTQIA cancer patients and caregivers	Mixed-methods	ADAPT-ITT framework. Focus groups, literature review, feedback sessions. The participants received training in research methods through an adapted version of the Tufts Clinical and Translational Science Institute curriculum, "Building Your Capacity (BYC): Advancing Research through Community Engagement" (	In lieu of a formal qualitative data analysis, notes from the review sessions were collated by members of the research team, and illustrative quotes were extracted.	n = 13 stakeholders (cancer survivors, patients, family caregivers, advocates)	*Facilitators:* Training in research methods *Challenges:* Not reported	Adapted intervention for LGBTQIA populations	Increased confidence in engaging with researchers

(*continued on next page*)

Table 2 (*continued*)

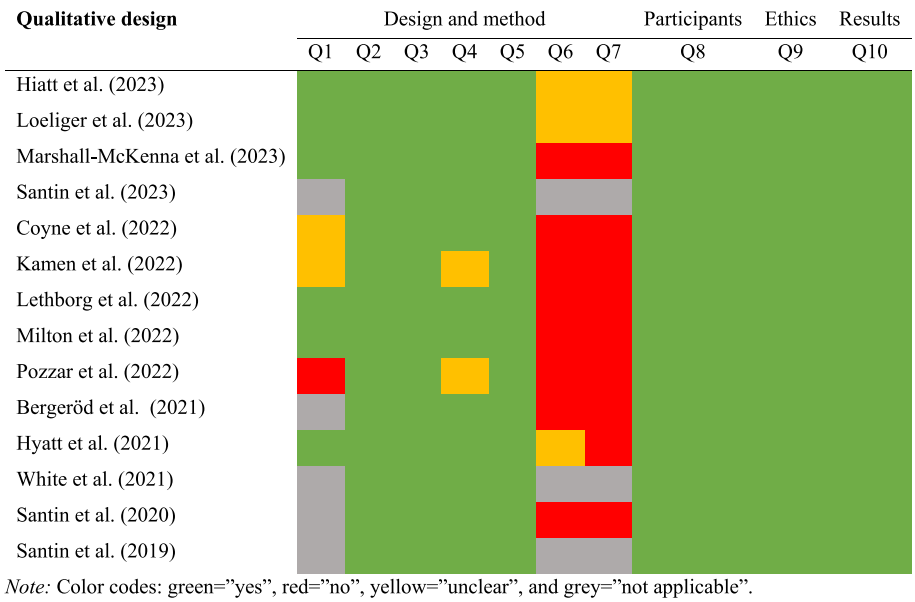
First author, publication year, location	Aims	Study type	Co-design approach, methods, and preparation of stakeholders	Analysis	Stakeholder characteristics	Co-design facilitators and challenges	Intervention	Key outcomes
			Rubin et al., 2012).					
Lethborg et al. (2022), Australia	To develop, implement, and evaluate a culturally informed and co-designed supportive care model for Aboriginal and Torres Strait Islander people with cancer and their families	Mixed-methods	Appreciative Inquiry (AI). Literature review, clinical data audit, advisory group meetings, feedback sessions. Preparation involved building and strengthening relationships with Aboriginal and Torres Strait Islander people.	A data extraction tool was used to review medical records and assess intervention impact, with a complexity score based on stressors and strengths.	Aboriginal Liaison Officers, social workers, community members	*Facilitators:* Strong community relationships. *Challenges:* Limited formal feedback, small participant number	Person-centered, culturally sensitive care model	Practical, financial, and emotional support
Milton et al. (2022), UK	To develop a social intervention to support people affected by a cancer diagnosis to be physically active	Mixed-methods	Framework for the Co-production and Prototyping of Public Health Interventions. Semi-structured interviews, workshops	Inductive content analysis	n = 11 patients, n = 7 family caregivers, n = 6 healthcare professionals, n = 8 physical activity providers, n = 5 cancer charity representatives	*Facilitators:* High stakeholder engagement *Challenges:* Not reported	"All Together Active" physical activity intervention	Increased likelihood of adoption by target populations
Pozzar et al. (2022), USA	To develop and field test the BOLSTER intervention to support patients and family caregivers after hospitalizations for peritoneal carcinomatosis	Mixed-methods	ADAPT-ITT framework. Interviews, feedback from advisory council, expert verification.	Not specified	Patients, family caregivers, clinicians	*Facilitators:* Advisory council involvement *Challenges:* Not reported	BOLSTER intervention	Comprehensive support for complex care needs
Bergerød et al. (2021), Norway	To present the results from a consensus-based participatory process of designing a guide for next-of-kin involvement in hospital cancer care	Qualitative	Consensus process based on modified nominal group technique. Consensus meeting, post-feedback. Preparation of stakeholders consisted of reading activities and reflective writing.	Content analysis	n = 20 participants (next-of-kin, oncology nurses, physicians)	*Facilitators:* Proper preparation, safe discussion environment *Challenges:* Need for more next-of-kin representatives	Guide for next-of-kin involvement	Improved involvement in hospital cancer care
Hyatt et al. (2021), Australia	To use a co-design process to develop video resources about immunotherapy for patients and family carers	Qualitative	Experience-based co-design (EBCD). Qualitative interviews, workshop, feedback and editing.	Not specified	n = 23 patients, n = 8 family caregivers, n = 9 clinicians, n = 3 consumer advocates	*Facilitators:* Relevant topic identification *Challenges:* Not reported	Video resources about immunotherapy	Targeted information for immunotherapy patients
White et al. (2021), Australia	To develop and trial the Cancer Readiness Tool (CRT)	Qualitatve	Co-design process. Literature review, environmental scan, interviews.	Thematic analysis	n = 47 adult family caregivers	*Facilitators:* Not reported *Challenges:* Not reported	Cancer Readiness Tool	Cancer capacity assessment tool
Santin et al. (2020), Vietnam	To develop a web-based resource to provide information and pscyhosocial support to informal cancer carers in hospitals in Vietnam	Qualitative	Co-design methodology. Interviews, focus groups, verification events.	Thematic analysis	n = 23 healthcare professionals, n = 20 informal cancer carers	*Facilitators:* Not reported *Challenges:* Not reported	Web-based resource for informal cancer carers	Information and psychosocial support
Santin et al. (2019), UK	To co-design and test the acceptability of a	Qualitative	Six-step co-design model. Workshops,	Thematic analysis, descriptive	n = 8 cancer carers, n = 12	*Facilitators:* Positive response to	Peer-led web-based resource (PLWR)	Practical and emotional advice

(*continued on next page*)

Table 2 (*continued*)

First author, publication year, location	Aims	Study type	Co-design approach, methods, and preparation of stakeholders	Analysis	Stakeholder characteristics	Co-design facilitators and challenges	Intervention	Key outcomes
	peer-led web-based resource (PLWR) for cancer caregivers		expert advisory team meetings, user testing.	statistics, Google analytics	expert advisory team members	resource *Challenges:* Homogenous sample of family caregivers		for family caregivers

Table 3
| Critical appraisal of included studies.



Note: Color codes: green="yes", red="no", yellow="unclear", and grey="not applicable".

process (Milton et al., 2022).

The diversity of stakeholder involvement was a key factor in the co-design process. Engaging various stakeholders, including patients, family caregivers, healthcare professionals, and community members, ensured that the interventions were comprehensive and addressed the needs of all relevant groups. For example, the study by Lethborg et al. involved Aboriginal Liaison Officers, social workers, and representatives from Aboriginal community-controlled health services, which helped in developing a culturally informed supportive care model for Aboriginal and Torres Strait Islander people with cancer. Furthermore, the iterative nature of the co-design process facilitated continuous improvement and refinement of interventions (Lethborg et al., 2022). Santin et al. used a six-step co-design model that supported rapid development and iterative feedback, ensuring that the final product was well-aligned with the needs of family caregivers and healthcare professionals (Santin et al., 2023).

Training and capacity-building for participants also contributed to successful co-design. Kamen et al. reported that providing research methods training to members of the LGBT Cancer Action Council increased their confidence and ability to engage effectively in the co-design process (Kamen et al., 2022). This training helped bridge the gap between academic researchers and community stakeholders, fostering a more collaborative and productive environment.

4.3.3. Challenges in the co-design process

Despite the numerous co-design facilitators, several challenges were consistently reported across the included studies, such as recruitment difficulties, cultural and contextual barriers, time constraints, and scheduling issues. A significant problem was recruitment, especially in engaging certain participant groups such as family caregivers. Santin et al. highlighted substantial challenges in recruiting family caregivers, resulting in more healthcare professionals participating in the co-design process (Santin et al., 2023). Similarly, Marshall-McKenna et al. found it challenging to engage family caregivers, affecting the diversity of perspectives in their study (Marshall-McKenna et al., 2023). Loeliger et al. also highlighted that healthcare professionals comprised the majority of workshop participants, which could bias results toward their perspectives over those of patients or family caregivers (Loeliger et al., 2023).

Further, cultural and contextual barriers were identified, particularly in studies with diverse populations. Lethborg et al. emphasized the importance of building and maintaining trust with Aboriginal and Torres Strait Islander communities, which required significant time and effort (Lethborg et al., 2022). Furthermore, Kamen et al. highlighted the need for cultural sensitivity and adapting interventions to meet the specific needs of LGBTQIA + cancer patients and their family caregivers (Kamen et al., 2022).

Time constraints and scheduling issues also presented significant challenges. Loeliger et al. reported that the limited time available for co-design workshops constrained the depth and breadth of discussions, with only two workshops conducted instead of the ideal three or four (Loeliger et al., 2023). This limitation was also apparent in the study by Hiatt et al., who found it challenging to schedule workshops at

convenient times for all participants, particularly those with work commitments (Hiatt et al., 2023).

4.3.4. Outcomes of the co-design process

The co-design process in the included studies generated several positive outcomes, significantly enhancing the relevance and effectiveness of the developed interventions. Outcomes reported were the creation of tailored interventions that directly addressed the specific needs and preferences of families affected by cancer. For instance, the CanEAT pathway provided centralized resources and tools for cancer nutrition care, empowering and promoting self-management strategies for patients and family caregivers and enhancing overall family health (Loeliger et al., 2023). Similarly, the study by Hiatt et al. developed a framework for improving access to nutrition information and support for head and neck cancer patients and their family caregivers (Hiatt et al., 2023). By involving patients, family caregivers, and healthcare professionals in the co-design process, the framework was tailored to meet the unique needs of this population, enhancing its relevance and usability. Another study created video resources for family caregivers of brain and general cancer patients, offering essential health information to help family caregivers manage their roles more effectively and reduce their burden (Coyne et al., 2022).

The interventions developed through co-design also aimed to support family health by addressing psychosocial needs. A peer-led web-based resource offered practical and emotional advice to family caregivers, fostering a supportive community and reducing feelings of isolation (Santin et al., 2019). Furthermore, the culturally informed supportive care model for Aboriginal and Torres Strait Islander people with cancer emphasized the importance of cultural sensitivity and community strengths, promoting holistic family health (Lethborg et al., 2022).

The BOLSTER intervention provided comprehensive support for patients and family caregivers following hospitalization for gynecologic cancer (Pozzar et al., 2022). This intervention included illustrated learning modules, written materials, and a smartphone application to guide self-management and monitor patient-reported outcomes, thereby supporting physical and emotional health. The digital component was also apparent in the LifeChamps platform, which aimed to enhance supportive cancer care for older adult cancer survivors and their families by addressing their healthcare needs, preferences, and expectations (Marshall-McKenna et al., 2023).

5. Discussion

To our knowledge, this study is the only scoping review of evidence related to current co-design approaches and activities in family interventions in cancer.

5.1. Search results

The screening process for this scoping review was extensive, with 3024 references screened for eligibility. Fourteen studies met the inclusion criteria and were included in the review. Although the studies were predominantly from Western countries, the studies were geographically diverse, highlighting the global interest in co-designing family interventions for cancer care.

The included studies varied in their participant groups, reflecting a broad range of perspectives (e.g., patients, family caregivers, healthcare professionals, patient advocates, charity representatives, consumer advocates, and community members). This variation underscores the importance of involving multiple stakeholders in the co-design process to ensure that interventions are comprehensive and address the needs of all relevant stakeholders. However, one study had missing data on co-design participants, and four studies did not specify the number or type of stakeholders participating in the co-design process. These gaps highlight the need for more detailed reporting in future research to enhance transparency.

5.2. Critical appraisal

The omissions related to not addressing the relationship between researchers and participants and the researcher's influence on the research, suggest a need for more transparency in reporting how researchers' perspectives and interactions may have impacted the study findings. The consistent failure to address key appraisal questions highlights the need for improved reporting standards and methodological rigor in future research for co-design studies. Appropriate reporting standards enhance co-design studies' transparency, reliability, and credibility. The call for improved reporting standards is echoed by Pearce et al., who argue that the iterative nature of co-design processes can resolve methodological challenges but requires a commitment to transparency in reporting (Pearce et al., 2022).

5.3. Study themes

The following themes will be discussed: Co-design approach and frameworks, Facilitators in the co-design process, Challenges in the co-design process, and Outcomes of the co-design process.

Various frameworks and approaches can be helpful when co-designing family interventions in cancer care. The most commonly used co-design framework among the included studies was the Experience-based co-design, applied in four studies (Coyne et al., 2022; Hiatt et al., 2023; Hyatt et al., 2021; Loeliger et al., 2023). Using a well-established co-design framework and toolkit was considered crucial for structuring the co-design process and ensuring that the final product aligns with stakeholder needs (Hiatt et al., 2023; Santin et al., 2023). Several studies provide evidence supporting this assertion. Yardley et al. highlight the importance of adopting a structured framework to enhance the acceptability and feasibility of interventions (Yardley et al., 2015). They argue that synthesizing evidence from both qualitative and quantitative research is crucial for identifying key issues and challenges that interventions must address, ensuring the design process is driven by stakeholder needs. Murfield et al. further illustrate this point by demonstrating how a co-design approach centered on the needs and preferences of family caregivers of individuals with dementia led to a more relevant and acceptable intervention (Murfield et al., 2022). They highlighted that without in-depth co-design work, critical issues might remain unaddressed, underscoring the necessity of a structured approach to capture stakeholder insights effectively (Murfield et al., 2022).

Despite a recognition of its importance, engaging family caregivers in co-designing proved to be a challenge. This scoping review reported difficulties in recruiting family caregivers across the included studies. This difficulty can lead to potential biases in the results, favoring the perspectives of other stakeholders (Loeliger et al., 2023; Marshall-McKenna et al., 2023; Santin et al., 2023). The complexity of integrating diverse stakeholder perspectives in co-design is discussed by Grynne et al., noting that the process can be fraught with challenges that may lead to the overshadowing of family caregiver input (Grynne et al., 2021) like the overshadowing of family caregivers' needs that may occur in clinical cancer care (Samuelsson et al., 2022). Similarly, Göttgens and Oertelt-Prigione emphasize that varying motivations and priorities among stakeholders can complicate the co-creative space, potentially marginalizing the voices of family caregivers (Göttgens and Oertelt-Prigione, 2021). However, Wittenberg et al. emphasizes that family caregivers often possess differing levels of health literacy, affecting their ability to communicate their needs and engage effectively in the care process (Wittenberg et al., 2019). This variation necessitates tailored support to ensure family caregivers can articulate their experiences and requirements, thereby preventing their needs from being overshadowed by other stakeholders.

Moreover, family caregivers' emotional and psychological burdens

can further complicate their engagement in co-design processes. Meeker et al. highlight that family caregivers often encounter barriers related to their knowledge, beliefs, and attitudes in cancer care, hindering their ability to participate fully in discussions about care strategies (Meeker et al., 2011). This lack of confidence and knowledge can lead to feelings of inadequacy, as Hussin and Sabri noted, where family caregivers experience guilt and stress related to their perceived competence in providing care (Mohamed Hussin and Mohd Sabri, 2023). Such emotional challenges can inhibit family caregivers from voicing their needs (Samuelsson et al., 2024), ultimately biasing the co-design outcomes.

Cultural and contextual barriers may impact the co-design of family interventions in cancer care, particularly among diverse populations, necessitating the need for culturally sensitive approaches to health interventions (Lethborg et al., 2022). Iwelunmor et al. underscores the necessity of building community trust, highlighting how cultural responsibilities can influence health behaviors (Iwelunmor et al., 2014). Such cultural dynamics necessitate that interventions are designed with cultural sensitivity and adapted to the specific community context of the population they aim to serve. Moreover, socioeconomic status plays a critical role in who participates in cancer research and interventions. Schuster et al. note that racial and ethnic diversity remains insufficient in cancer genomics research, exacerbating disparities in cancer care for underrepresented populations, including Black, Hispanic, and Indigenous communities (Schuster et al., 2023). This lack of representation can lead to interventions that do not adequately address the unique needs of these groups.

The co-design process in family interventions in cancer care has demonstrated positive outcomes, enhancing the relevance and effectiveness of the developed strategies. The creation of tailored interventions that address the specific needs and preferences of families affected by cancer is an outcome of this process. For instance, nutritional interventions that empower families with the knowledge, skills, and support necessary to manage nutritional care can promote self-management strategies and overall family health (Hiatt et al., 2023; Loeliger et al., 2023). Such empowerment is crucial, as studies indicate that effective nutritional interventions can significantly improve patients with cancer and their family caregivers' quality of life (Liposits et al., 2023). Another outcome of a co-design process was the development of video resources for family caregivers (Coyne et al., 2022). Such resources provide essential health information that helps family caregivers manage their roles more effectively and reduce their burden (Dwyer et al., 2020). The interventions developed through co-design also addressed psychosocial needs, which are critical for supporting family health. An example is the peer-led web-based resource offering practical and emotional advice to family caregivers (Santin et al., 2019), fostering a supportive community and reducing feelings of isolation. This aligns with findings that emphasize the importance of social support in enhancing the well-being of caregivers, as they often experience significant emotional and psychological stress (Gabriel et al., 2022). In conclusion, the co-design process has led to the creation of tailored resources that effectively address the diverse needs of patients with cancer and their family caregivers.

Across the included studies, co-design activities were reported to enhance the relevance, usability, and acceptability of the developed interventions. For example, nutritional tools, digital resources, and caregiver-focused materials were better tailored to user needs through stakeholder involvement (Hiatt et al., 2023; Loeliger et al., 2023; Santin et al., 2019). These outcomes suggest that co-design can strengthen alignment between intervention content and end-user preferences. However, several challenges limited the full potential of co-design. Recruitment of family caregivers was often difficult, leading to underrepresentation of their perspectives (Santin et al., 2023; Loeliger et al., 2023). Additionally, cultural, contextual, and logistical barriers affected participation across diverse populations. While structured frameworks supported the co-design process, few studies formally evaluated its impact on implementation or clinical outcomes, highlighting a need for more systematic assessment in future research.

5.4. Limitations

While comprehensive, the search terms of this scoping review may present some limitations. Only 14 studies were identified, which may indicate that the field of research co-design in family interventions in oncology is still underdeveloped. However, this limited number may also be influenced by the inclusion criteria and/or the search strategy applied. Specifically, the review included only studies involving adult patients and adult family members, thereby excluding studies focusing on children as either patients or family caregivers. This decision was based on the assumption that the challenges and needs related to co-design may differ when children are involved. Nevertheless, it is acknowledged that co-design frameworks could be applicable across age groups. Future reviews may consider including pediatric settings to capture a broader scope of co-design approaches.

Including various synonyms and related terms, although intended to capture all relevant literature, may lead to redundancy and overlap. Furthermore, focusing on English-language terms may introduce language bias, potentially overlooking research published in Danish, Norwegian, Spanish, and Swedish. Moreover, the complexity and length of the search string could pose challenges in replicating the search, affecting the reproducibility of this review.

Another limitation of this review is using the JBI Critical Appraisal Tool for Qualitative Research to appraise mixed-methods studies. While this tool effectively assesses the qualitative components, it may not fully capture the quality aspects of the quantitative components of these studies. Ideally, both qualitative and quantitative components should be appraised using appropriate tools.

6. Conclusion

Overall, the co-design processes facilitated the development of relevant interventions to address the multifaceted needs of families affected by cancer. These interventions aimed to promote, maintain, and restore family health by providing tailored support, enhancing self-management capabilities, and fostering a supportive community environment. Various co-design approaches were utilized, including Experience-Based Co-design and the ADAPT-ITT framework. Among the most commonly used co-design activities were interviews, workshops, focus groups, and feedback sessions, demonstrating a strong emphasis on iterative engagement with stakeholders. Key success factors of these interventions were a high degree of participant engagement, structured frameworks, a diversity of stakeholders, and training and capacity-building. Identified challenges that require consideration in planning a family intervention in cancer care concern: recruitment difficulties, cultural and contextual barriers, time constraints, and scheduling issues. Future research may explore methods to overcome these challenges. Enhanced reporting standards may improve co-design studies transparency, reliability, and credibility.

CRediT authorship contribution statement

Camilla S. Rothausen: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cristina Alfaro-Diaz:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Maria Samuelsson:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Registration and protocol

Registration can be accessed from the Open Science Framework

(https://doi.org/10.17605/OSF.IO/2RWB9) and the protocol at (https://doi.org/10.1136/bmjopen-2024-093961).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejon.2025.102966.

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